2.1 Equations of Motion

Question Paper

Course	CIE A Level Physics
Section	2. Kinematics
Topic	2.1 Equations of Motion
Difficulty	Easy

Time allowed: 50

Score: /37

Percentage: /100

Question 1a

State the difference between

(i)
distance and displacement,

[2]

(ii)
speed and velocity.

[2]

[4 marks]

Question 1b

Complete the table in Table 1.1 to describe the features of displacement-time, velocity-time and acceleration-time graphs represent.

Place a cross (✘) in any grid space which does not represent anything. Some grid spaces have been completed for you.

Table 1.1

	displacement-time	**velocity-time**	**acceleration-time**
gradient represents			✘
area under curve represents			
y-intercept represents			
horizontal line represents	zero velocity		
straight section represents			
curved section represents		non-uniform acceleration	

[6 marks]

Question 1c

On Fig. 1.2, sketch the variation of time with displacement, velocity, and acceleration for an object moving with **constant velocity**.

Take the initial displacement as zero.

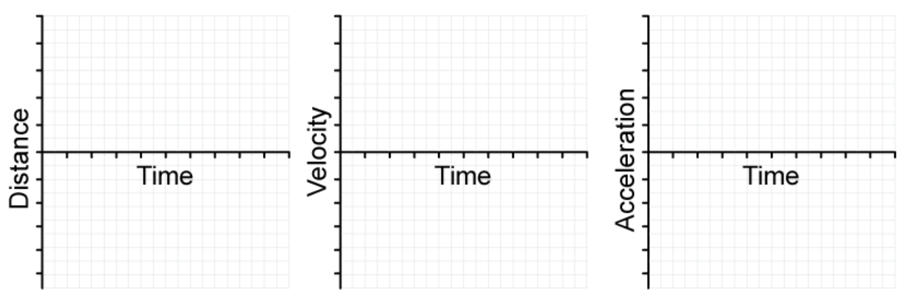


Fig. 1.2

[3 marks]

Question 1d

On Fig. 1.3, sketch the variation of time with displacement, velocity, and acceleration for an object moving with **uniform acceleration**.

Take the initial displacement as equal to zero.

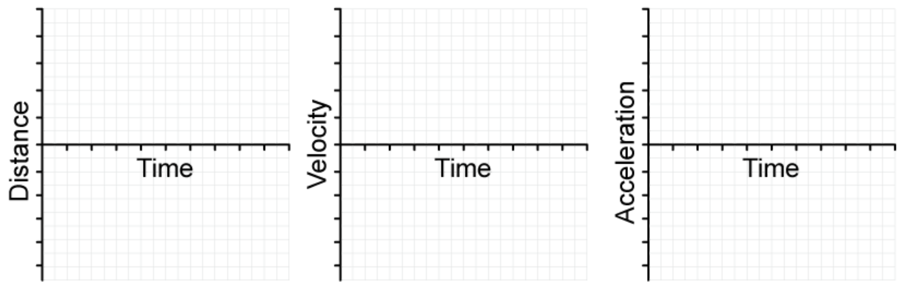


Fig. 1.3

[3 marks]

Question 2a

Place ticks (✓) in the table in Table 1.1 to indicate which of the quantities are vectors and which are scalars.

Table 1.1

	velocity	speed	distance	displacement	acceleration
vector					
scalar					

[2 marks]

Question 2b

Two of the equations of uniformly accelerated motion are

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

State

(i)
the definition of uniform acceleration,

[1]

(ii)
the **two** other equations of motion that can be used to derive the above equations.

[2]

[3 marks]

Question 2c

Fig. 1.2 shows an electric motor which is used to lift and lower heavy loads.

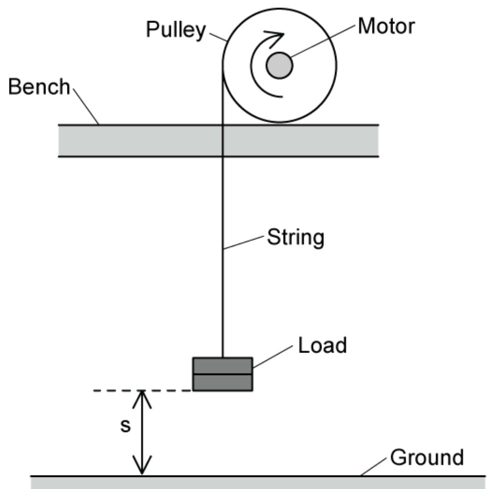


Fig. 1.2

Initially, the load is on the ground. Fig. 1.3 shows the variation of the displacement *s* of the load with time *t*.

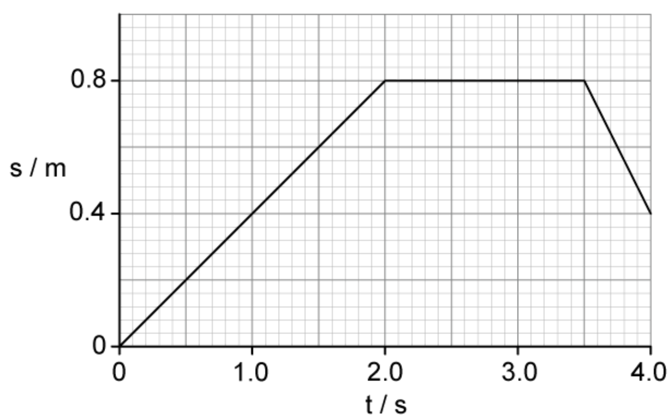


Fig. 1.3

(i)
Complete Table 1.4 to show the magnitude and direction of the velocity of the load during the time intervals shown.

Table 1.4

time interval	magnitude of velocity	direction of velocity
t = 0 to 2.0 s		
t = 2.0 to 3.5 s		
t = 3.5 to 4.0 s		

[3]

(ii)

On Fig. 1.5, sketch a graph to show the variation of the velocity *v* of the load with time *t*.

Include values for velocity on the *v*-axis.

[2]

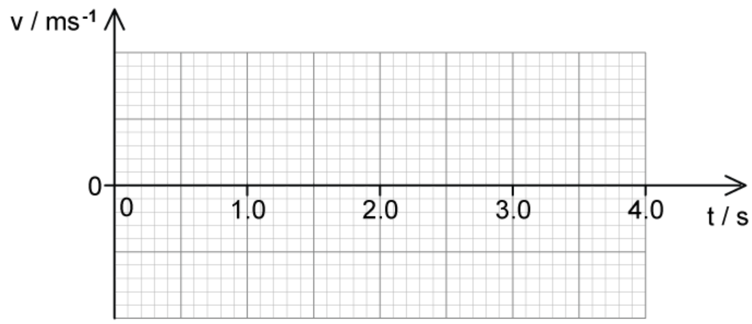


Fig. 1.5

[5 marks]

Question 2d

As the load is lowered towards the ground, the string breaks when t = 4.0 s. It then falls vertically toward the ground.

Calculate the velocity of the load just before it hits the ground.

Assume air resistance is negligible.

[2 marks]

Question 3a

A student drops a stone into a well and measures the time it takes to hear the stone splash at the bottom.

Explain how this measurement of time can be used to find the depth of the well.

[3 marks]

Question 3b

It takes 3.2 s for the stone to drop from rest and splash into the water at the bottom.

Calculate the speed of the stone when it hits the water.

[2 marks]

Question 3c

On Fig. 1.1, sketch the variation of time with velocity for the stone as it falls down the well.

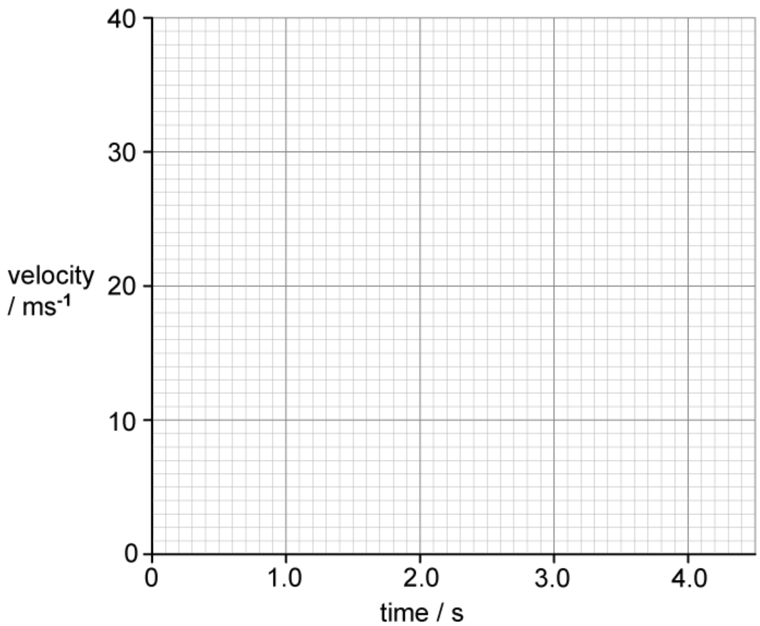


Fig. 1.1

[2 marks]

Question 3d

Use your graph from **(c)** to determine the depth of the well.

[2 marks]

Model Answers: Easy

1a

(a)

(i) The difference between distance and displacement is:

- Distance is defined as the total length between two points **OR** distance is a scalar quantity; [1 mark]
- Displacement is defined as the distance from a (fixed) point in a specified direction **OR** displacement is a vector quantity; [1 mark]

(ii) The difference between speed and velocity is:

- Speed is defined as the total distance travelled per unit of time **OR** speed is a scalar quantity; [1 mark]
- Velocity is defined as the rate of change of displacement of an object **OR** velocity is a vector quantity; [1 mark]

[Total: 4 marks]

1b

(b) The correctly complected table is as follows:

	displacement-time	**velocity-time**	**acceleration-time**
gradient represents	**velocity**	**acceleration**	✘
area under curve represents	✘	**change in displacement**	**change in velocity**
y-intercept represents	**initial displacement**	**initial velocity**	**initial acceleration**
horizontal line represents	zero velocity	**constant velocity**	**constant / uniform acceleration**

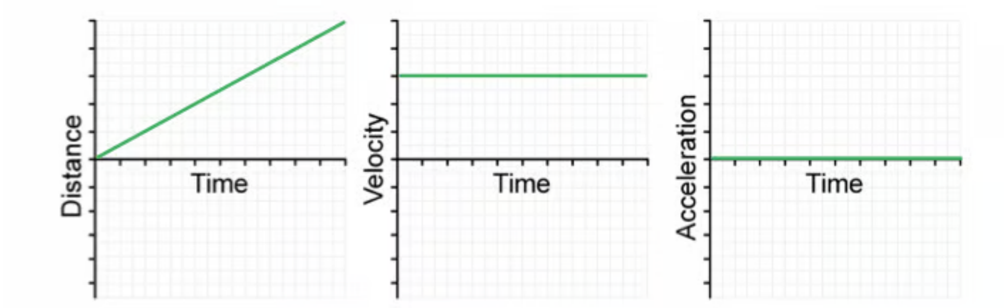
straight section represents	**constant velocity**	**constant / uniform acceleration**	✘
curved section represents	**changing velocity / acceleration**	non-uniform acceleration	✘

- *One mark for each correct row*

[Total: 6 marks]

1c

(c) For an object moving with constant velocity:

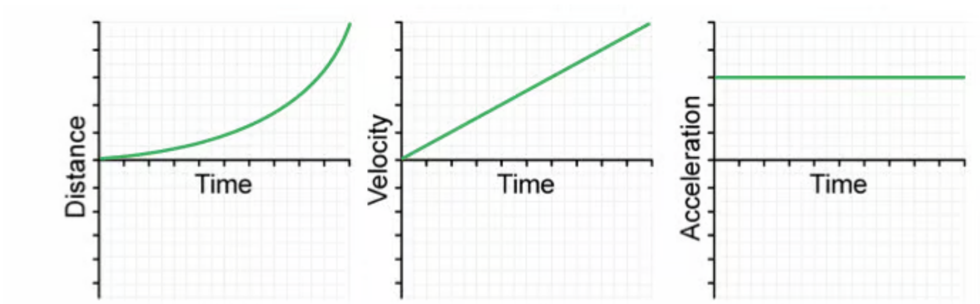


- Displacement-time graph: straight, diagonal line starting from zero drawn; [1 mark]
- Velocity-time graph: straight, horizontal non-zero line drawn; [1 mark]
- Acceleration-time graph: straight, horizontal drawn at zero; [1 mark]

[Total: 3 marks]

1d

(d) For an object moving with uniform acceleration:



- Displacement-time graph: curved / parabolic line starting from zero drawn; [1

mark]
- Velocity-time graph: straight, diagonal line starting from zero drawn; [1 mark]
- Acceleration-time graph: straight, horizontal non-zero line drawn; [1 mark]

[Total: 3 marks]

2a

(a) A correctly completed table is as follows:

	velocity	speed	distance	displacement	acceleration
vector	✓			✓	✓
scalar		✓	✓		

- Vectors: velocity, displacement, acceleration; [1 mark]
- Scalar: speed, distance; [1 mark]

[Total: 2 marks]

2b

(b)

(i) Uniform acceleration is defined as:

Any **one** from:

- When an object's velocity increases (or decreases) at a constant rate; [1 mark]
- Acceleration in which the rate of change of velocity is constant; [1 mark]
- When the velocity of an object changes by equal amounts in equal time periods; [1 mark]

(ii) The two other equations of motion are:

- $v = u + at$ [1 mark]
- $s = \frac{(u + v)}{2}t$ [1 mark]

[Total: 3 marks]

2c

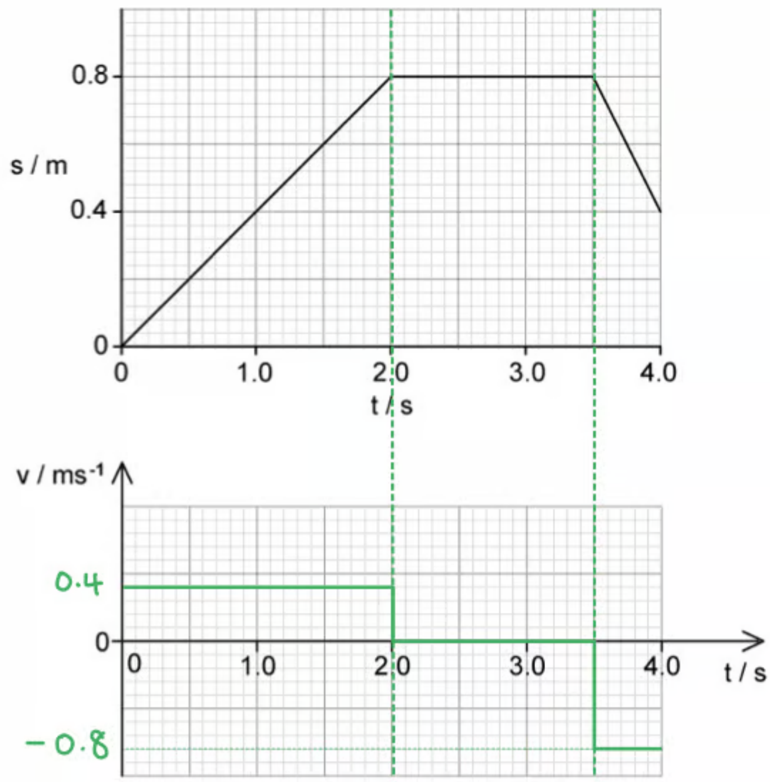
(c)

(i) The variation of the velocity of the load from $t = 0$ to 4.0 s is:

time interval	magnitude of velocity	direction of velocity
$t = 0$ to 2.0 s	**0.4 m s^{-1}**	**upwards / positive**
$t = 2.0$ to 3.5 s	**0**	**none**
$t = 3.5$ to 4.0 s	**0.8 m s^{-1}**	**downwards / negative**

- *One mark for each correct row*

(ii) The velocity-time graph is as follows:



- All lines drawn straight and horizontal; [1 mark]

- Values +0.4 m s^{-1} and –0.8 m s^{-1} in correct places on the v-axis; [1 mark]

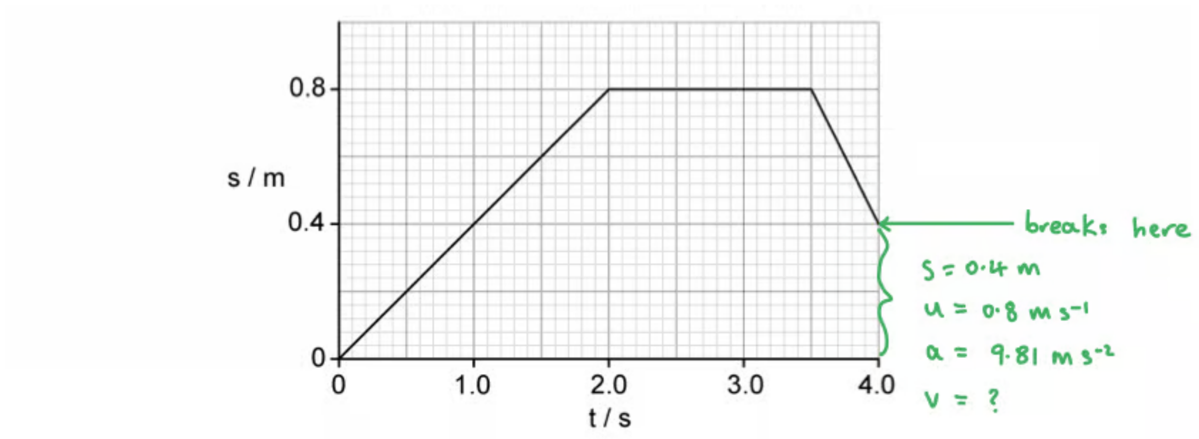
Example of calculation:

- Upwards velocity: $v = \frac{\Delta s}{\Delta t} = \frac{0.8}{2.0} = 0.4\ m\ s^{-1}$
- Downwards velocity: $v = \frac{\Delta s}{\Delta t} = \frac{0.4 - 0.8}{4.0 - 3.5} = -0.8\ m\ s^{-1}$

[Total: 5 marks]

2d

(d) Calculate the velocity of the load just before it hits the ground:



List the known quantities, now defining downwards as the positive direction (u is now positive):

- Initial velocity, u = 0.80 m s^{-1} (from **(c)**)
- Acceleration, $a = g$ = 9.81 m s^{-2}
- Distance, s = 0.40 m
- Final velocity = v

Select the correct SUVAT equation:

- $v^2 = u^2 + 2as$

Substitute the values and evaluate:

- $v^2 = 0.80^2 + (2 \times 9.81 \times 0.40)$ [1 mark]

- $v = \sqrt{8.488}$
- v = 2.913 = 2.9 m s^{-1} (2 s.f.) [1 mark]

[Total: 2 marks]

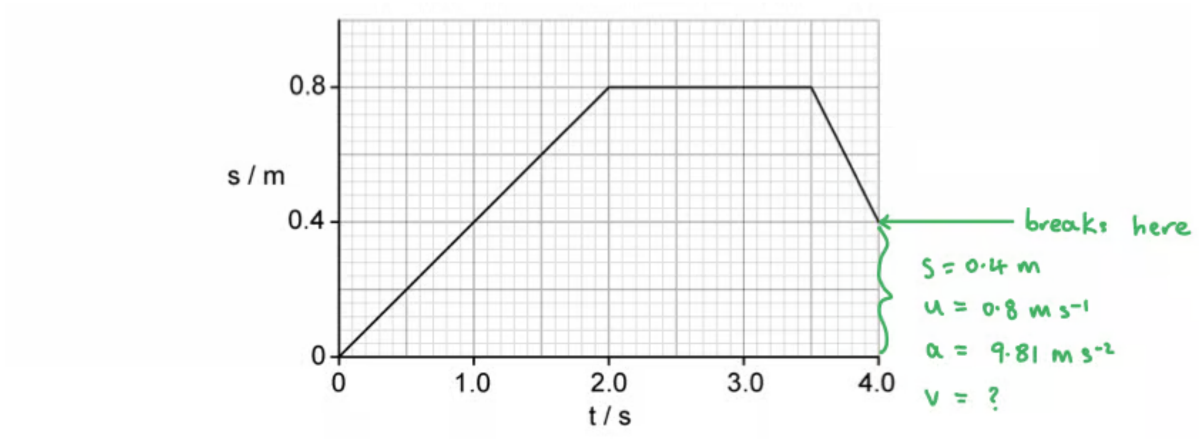
Make sure to use the correct initial velocity here - the 0.40 m s^{-1} was the velocity as the load was being lifted **up** (not down, as this question is asking about.)

For questions like this, you can choose which direction is the positive direction, just be **clear** which you are doing in your calculations and don't mix your directions up.
In this question, if we were going to keep **up** as the positive direction, our values would be:

- Initial velocity, u = –0.80 m s^{-1}
- Acceleration, a = g = –9.81 m s^{-2}
- Distance, s = –0.40 m

Distance is covered downwards, acceleration acts downwards and initial velocity is downwards - all in the negative direction. It can be easier to just say "downwards is the positive direction" and avoid all those minus signs.

(d) Calculate the velocity of the load just before it hits the ground:



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3a

(a) This measurement of time can be used to find the depth of the well because:

- The initial speed *u* is zero **AND** the acceleration (due to gravity) is g = 9.81 m s^{-2} [1 mark]
- Final speed *v* can be found from $v = u + at$ **OR** $a = \frac{\Delta v}{t}$ [1 mark]
- The depth *s* can be found using $v^2 = u^2 + 2as$ and $s = ut + \frac{1}{2}at^2$ **OR** from the area under a graph of *v* against *t*; [1 mark]

[Total: 3 marks]

3b

(b) To calculate the speed of the stone when it hits the water:

List the known quantities:

- Time, t = 3.2 s
- Acceleration, g = 9.81 m s^{-1}
- Initial velocity, u = 0

Calculate the final speed *v*:

- $v = u + at$
- v = 0 + (9.81 × 3.2) [1 mark]
- v = 31.392 = 31 m s^{-1} [1 mark]

[Total: 2 marks]

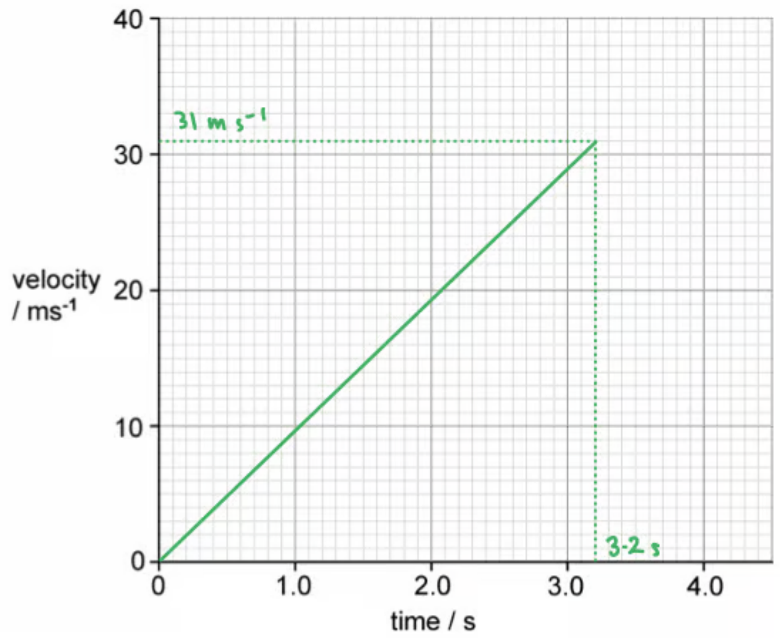
3c

(c) On Fig. 1.1, sketch the variation of time with velocity for the stone as it falls down the well.

List the known quantities:

- Final speed, $v = 31 \text{ m s}^{-1}$ (from part **(b)**)
- Time taken, $t = 3.2$ s

The variation of time with velocity for the stone is as follows:

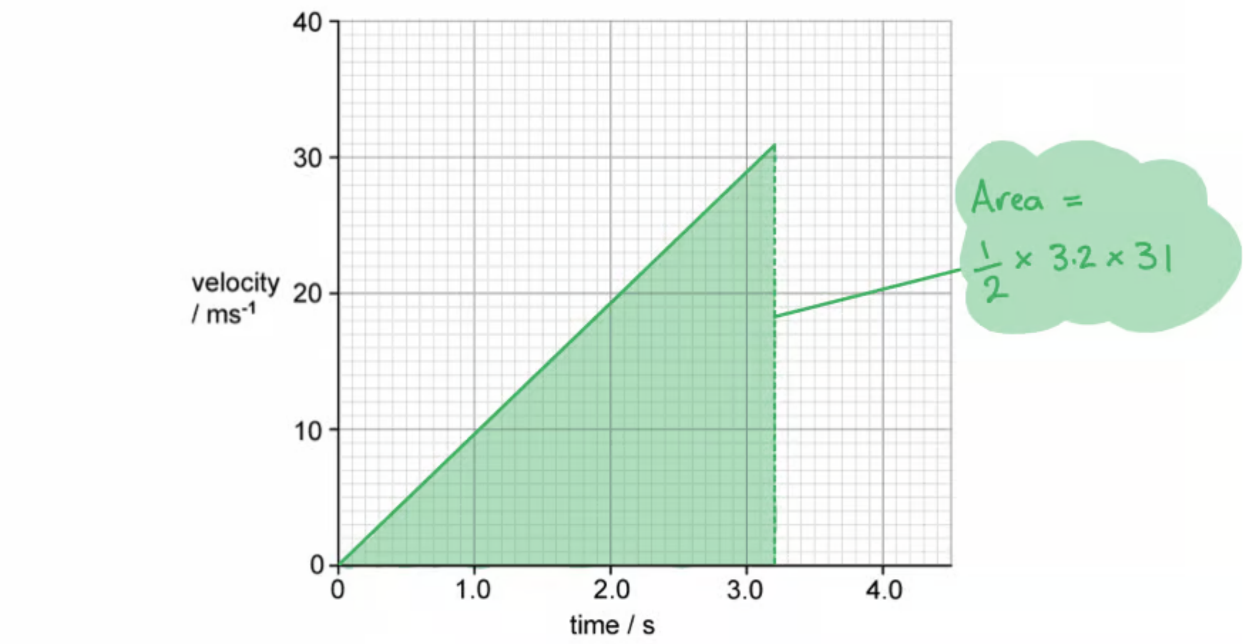


- Straight line drawn starting from zero; [1 mark]
- Line stops at $t = 3.2$ s, $v = 31 \text{ m s}^{-1}$; [1 mark]

[Total: 2 marks]

3d

(d) Determine the depth of the well from the graph:



- Distance travelled = area under the velocity-time graph
- Depth = $\frac{1}{2} \times 3.2 \times 31$ [1 mark]
- Depth = 49.6 = 50 m [1 mark]